

20/7/2026

Assessment - 2

12. Design a machine learning model for spam e-mail classification using labeled data.
- Explain the choice of algorithm
 - Describe the pre-processing steps such as tokenization, stop word removal, and vectorization.
 - Discuss evaluation metrics such as accuracy, precision, recall and f1-score.

Ans (a) Choice of Algorithm

A suitable algorithm for spam email classification is Naive Bayes because it is simple, fast and performed well on text classification problems.

Why Naive Bayes?

- Assumes that the occurrence of each word is independent of others.
- Works efficiently with large text datasets.
- Handles high-dimensional features created from text.
- Requires less training time compared to complex models.

Other Algorithms:

- Support Vector Machine (SVM): Provides high accuracy but requires more computation for large datasets.
- Logistic Regression: Offers good performance, is easy to interpret and works well with text features.

Chosen Algorithm: multinomial Naïves Bayes.

(b) Pre-processing Steps:

Before training the model, the email text must be converted into a suitable numerical format.

(i) Data Collection:

→ Collect labeled emails.

→ Labels :

Spam (1)

Not Spam / Ham (0)

(ii) Tokenization

→ Split each e-mail into individual words.

Example:

E-mail: "Congratulations, you won a free Prize".

Tokens: ["Congratulations", "you", "won", "a", "free", "Prize", "."]

(iii) Convert to Lowercase:

Congratulations → congratulations

free → free

(iv) Stop-word Removal

Remove common words that carry little meaning

Example:

the, is, am, are, a, an, and, of, in

(v) Remove special characters and punctuation.

"Win!!! \$\$\$ Now"

↓

"Win Now"

(vi) Vectorization:

Convert text into numerical feature vector.

Common techniques:

→ Bag of Words (BoW)

→ TF-IDF

Examples:

Word	Free	Win	Meeting	Prize
E-mail 1	2	1	0	1
E-mail 2	0	0	2	0

(c) Evaluation Metrics

After training, evaluate the model using the following metrics:

(i) Accuracy:

Measures the overall percentage of correctly classified emails.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Meaning: Higher accuracy indicates better overall performance.

Precision:

Measures how many emails predicted as spam are actually spam.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Importance: High precision reduces false spam detections.

Recall: Measures how many actual spam emails are correctly detected.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Importance: High Recall ensures fewer spam emails reach the inbox.

F1-Score: The harmonic mean of Precision and recall.

$$F_1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

Importance: Useful when the dataset is imbalanced.

Design of machine Learning model for Spam e-mail classification using labeled data.

Labeled E-mail dataset



Text preprocessing



vectorization



Train Naive Bayes Model



Predict Spam or Ham.



Evaluate using Accuracy, Precision, Recall, and f1-score.

13. In the context of the candidate-elimination algorithm, consider a dataset with attributes: Sky, Air Temp, Humidity, Wind, water, forecast and target concept EnjoySport.

- Explain the initialization of S and G .
- Show how S and G change with each training example.
- Compare candidate-elimination with find- S -algorithm in terms of generalization capability.

Ans Consider the attributes Sky, AirTemp, Humidity, Wind, water, forecast

Training Dataset

Ex	Sky	AirTemp	Humidity	Wind	water	fore cast	Enjoy Sport
1	Sunny	Warm	Normal	Strong	warm	Some	Yes
2	Sunny	Warm	High	Strong	warm	same	Yes
3	Rainy	Cold	High	strong	Warm	change	No
4	Sunny	Warm	High	Strong	Cool	change	No

(a) Initialization of S (Specific Boundary) and G (General Boundary)

The candidate-elimination Algorithm maintains two boundaries of the version space:

1. Specific Boundary (S)

→ Represents the most specific hypothesis

→ Initially, it is the most restrictive hypothesis

Initialization:

$S_0 = \phi, \phi, \phi, \phi, \phi$

where, ϕ means no attribute value has been assigned yet.

2. General Boundary (G)

→ Represents the most general hypothesis.

→ Initially, it accepts every possible instance.

Initialization:

$G_0 = ?, ?, ?, ?, ?$

where ? means any value is accepted.

(b) Changes in S and G with Each Training Example.

Ex 1: (Positive)

Sunny, warm, normal, strong, warm, Same → Yes

Update S to match the first positive example.

$S_1 = \text{Sunny, warm, Normal, strong, warm, Same.}$

$G_1 = ?, ?, ?, ?, ?$

Ex 2: (Positive)

Sunny, warm, high, Strong, warm, Same → Yes

Compare with S :

Only Humidity differs.

Generalize humidity to:

$S_2 = \text{Sunny, warm, strong, warm, same.}$

$G_2 = ?, ?, ?, ?, ?$

Ex 3:

High, strong, warm, change → No

S already rejects this negative example.
Specialize G minimally to exclude the negative
example while remaining more general than
Possible G becomes:

$G_3 = \langle \text{sunny}, ?, ?, ?, ?, ? \rangle$

$?, \text{warm}, ?, ?, ?, ?$

$?, ?, ?, ?, ?, \text{same}$

These hypothesis exclude the negative instance
while covering all positive examples.

Ex 4: (Positive)

Sunny, warm, high, strong, cool, change $\rightarrow Y_1$

Generalize S :

water differs $\rightarrow ?$

forecast differs $\rightarrow ?$

$S_4 = \text{Sunny}, \text{warm}, ?, \text{strong}, ?, ?$

Remove hypotheses from G that do not cover
this positive example.

Remaining G :

$G_4 = \text{Sunny}, ?, ?, ?, ?, ?$

$?, \text{warm}, ?, ?, ?, ?$

final version space

Specific Boundary

$S = \langle \text{sunny}, \text{warm}, ?, \text{strong}, ?, ? \rangle$

General Boundary

$G = \text{Sunny}, ?, ?, ?, ?, ?$

$?, \text{warm}, ?, ?, ?, ?$

(c) Comparison of candidate Elimination and find-s algorithm.

feature	Candidate Elimination	find-s
→ Uses positive Examples	Yes	Yes
→ Uses Negative Examples	Yes	No
→ Maintain specific Boundary	Yes	Yes
→ Maintains general Boundary	Yes	No
→ finds Entire version Space	Yes	No
→ Handles multiple consistent hypotheses	Yes	No
→ Generalises Capability	High	limited
→ Computational cost	Higher	Lower.

The candidate - Elimination Algorithm is more powerful than the find-s - algorithm because it makes both the specific and general boundaries of the hypothesis space. find-s - learns only a single most specific hypothesis and cannot fully represent the comparative version space.